Exercise 1

In the lecture we convinced ourselves that all maps are well-defined. As an inclusion, the first map is clearly injective.

Exactness at $\mathcal{O}_k^\times$ is clear, since $\rho(u) = 1 \in (\mathcal{O}_k/\mathfrak{m})^\times$ by definition means exactly $u \equiv 1 \mod {}^\times\mathfrak{m}$.

Exactness at $(\mathcal{O}_k/\mathfrak{m})^\times$: For $u \in \mathcal{O}_k^\times$ we have $(u) = (1)$, so $(u)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. Conversely, let $\alpha \in k^\times$, $(\alpha, \mathfrak{m}_0) = 1$ and suppose $(\alpha)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. That is, there is some $\varepsilon \in \mathcal{O}_k^\times$ s.t. $\varepsilon\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$, and $\rho(\alpha) = \rho(\varepsilon^{-1}) \in \rho(\mathcal{O}_k^\times)$.

Exactness at $\mathrm{cl}_k(\mathfrak{m})$ is clear, since both kernel and image consist exactly of the classes of principal ideals coprime to $\mathfrak{m}$.

Finally, we show more generally, that if $\mathfrak{m} \mid \mathfrak{n}$, then the induced map $\mathrm{cl}_k(\mathfrak{n}) \to \mathrm{cl}_k(\mathfrak{m})$ is surjective: Let $\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) \in \mathrm{cl}_k(\mathfrak{m})$, for some ideal $\mathfrak{a} \in I_k(\mathfrak{m})$. By the approximation theorem, we find $\alpha \in k$ with $\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$ and $\nu_\mathfrak{p}(\alpha) = -\nu_\mathfrak{p}(\mathfrak{a})$ for all prime ideals $\mathfrak{p} \mid (\mathfrak{n}, \mathfrak{a})$. Then $(\alpha\mathfrak{a}, \mathfrak{n}) = 1$, and

$$\alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{n}) \mapsto \alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) = \mathfrak{a}\mathcal{P}_k(\mathfrak{m}).$$

Exercise 2

If $0 \to A \to B \to C \to 0$ is a short exact sequence of groups, then B is partitioned into $|C|$ cosets each of size $|A|$, hence $|B| = |A| \cdot |C|$. If $0 \to A \to B \to C \to D \to 0$ is exact, let $K = \ker(C \to D) = \mathrm{im}(A \to B)$. Then $0 \to A \to B \to K \to 0$ and $0 \to K \to C \to D \to 0$ are exact, and therefore

$$|A| \cdot |C| = |A| \cdot |K| \cdot |D| = |B| \cdot |D|.$$

(More generally, given any finite exact sequence, continuing in this way yields an equation of "alternating" cardinalities.)

Applying this to the exact sequence $0 \to \mathcal{O}_k^\times/(\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times) \to \ldots \to 0$ yields

$$[\mathcal{O}_k^\times : \mathcal{O}_k^\times \cap k_\mathfrak{m}^\times] \cdot |\,\mathrm{cl}_k(\mathfrak{m})| = |\,\mathrm{cl}_k\,| \cdot |(\mathcal{O}_k/\mathfrak{m})^\times|.$$

In particular, since the right hand side is finite, and groups cannot be empty, both cardinalities on the left hand-side are finite as well, and dividing by the first term yields the result.

Exercise 3

Let $k = \mathbb{Q}(\sqrt{d})$ and let $\tau_1 = \mathrm{id}, \tau_2 : \bar{\cdot}$ be the two embeddings associated to $\infty_{1/2}$. By Dirichlet, $\mathcal{O}_K^\times = \{\pm 1\} \times \varepsilon^\mathbb{Z}$, where wlog $\varepsilon > 1$. Clearly $-1 \not\equiv 1 \bmod {}^\times\mathfrak{m}$. From $\mathrm{N}_{k/\mathbb{Q}}(\varepsilon) = \varepsilon\bar{\varepsilon}$ we see $\mathrm{sgn}(\bar{\varepsilon}) = \mathrm{N}_{k/\mathbb{Q}}(\varepsilon)$. Hence if $\mathrm{N}_{k/Q}(\varepsilon) = 1$, then $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^\mathbb{Z}$ and the index in $\mathcal{O}_k^\times$ is 2. Otherwise, $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^{2\mathbb{Z}}$ and the index in $\mathcal{O}_k^\times$ is 4.

Further, since $\mathfrak{m}_0 = (1)$ we have $|(\mathcal{O}_k/\mathfrak{m})^\times| = |\{\pm 1\}^2| = 4$. The result now follows from exercise 2.

Exercise 1

We have

$$\frac{(\mathcal{O}_K/\mathfrak{f})^\times}{U} \cong \ker(\mathrm{cl}_k(\mathfrak{f}) \to \mathrm{cl}_k) \cong \ker(\mathrm{Gal}(k(\mathfrak{f})/k) \to \mathrm{Gal}(k(1)/k)) \cong \mathrm{Gal}(k(\mathfrak{f})/k(1)),$$

where the first isomorphism follows from last week's sheet, the central one is induced by the Artin map (which is functorial), and the last is Galois theory.

Exercise 2

Let $\mathfrak{m} := q\mathbb{Z}\infty$. Then we know $\mathbb{Q}(\mathfrak{m}) = \mathbb{Q}(\zeta_q)$ and hence the Artin map induces an isomorphism

$$\mathrm{cl}_{\mathbb{Q}}(\mathfrak{m}) \cong \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \cong (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}/(q-1)\mathbb{Z}.$$

Let $H \subseteq \mathrm{cl}_Q(\mathfrak{m})$ be the unique subgroup of index 2. Then the existence theorem yields a unique abelian subfield $\mathbb{Q}(\zeta_q)/K/\mathbb{Q}$ with $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{cl}_{\mathbb{Q}}(\mathfrak{m})/H$, and ν ramifies in K only if $\nu \mid \mathfrak{m}$. Now the first condition implies that $K/\mathbb{Q}$ is quadratic, say $K = \mathbb{Q}(\sqrt{d})$ for some squarefree $d \in \mathbb{Z}$. Let p be a prime divisor of d. Then p ramifies in K, hence $p \mid q$. Hence either $d = -1$, or $d = \pm q$. Now 2 isn't allowed to ramify, so $d = -1$ is out, and we need the sign so that $\pm q \equiv 1 \bmod 4$. Hence if $q \equiv 1 \bmod 4$, we have $\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\zeta_q)$, and if $q \equiv 3 \bmod 4$, then $\mathbb{Q}(\sqrt{-q}) \subseteq \mathbb{Q}(\zeta_q)$, hence

$$\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\sqrt{-q}, \sqrt{-1}) = \mathbb{Q}(\sqrt{-q})\mathbb{Q}(\zeta_4) \subseteq \mathbb{Q}(\zeta_q)\mathbb{Q}(\zeta_4) = \mathbb{Q}(\zeta_{4q}).$$

In both cases this implies the claim, since $\mathbb{Q}(\sqrt{q})$ is totally real.

Exercise 3

Let $m := [k : \mathbb{Q}_p]$. Then we showed in ANT that $k^\times \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^m$ for a p-power q and non-negative integer a. Hence

$$k^\times/(k^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \oplus \mathbb{Z}/(n, q-1)\mathbb{Z} \oplus \mathbb{Z}/(n, p^a)\mathbb{Z} \oplus (\mathbb{Z}_p/n\mathbb{Z}_p)^m.$$

The first three summands are clearly finite, hence it remains to consider $\mathbb{Z}_p/n\mathbb{Z}_p$. Write $n = p^v b$ with $v, b \geq 0$, $p \nmid b$. Then b is invertible in $\mathbb{Z}_p$, hence $\mathbb{Z}_p/n\mathbb{Z}_p \cong \mathbb{Z}_p/p^v\mathbb{Z}_p$, which is isomorphic to $\mathbb{Z}/p^v\mathbb{Z}$ by ANT. Putting everything together, we see that $(k^\times)^n$ has finite index in $k^\times$.

To see that $(k^\times)^n$ is open, it suffices to find an $N \in \mathbb{N}$ with $1 + \mathfrak{p}_k^N \subseteq (k^\times)^n$, then $(k^\times)^n = \bigcup_{\alpha \in (k^\times)^n} \alpha(1 + \mathfrak{p}_k^N)$ is a union of open sets, hence open. To do this, let $s > \frac{e}{p-1}$ and $N = s + v_k(n)$. Then

$$1 + \mathfrak{p}_k^N = \exp(n\mathfrak{p}_k^s) = \exp(\mathfrak{p}_k^s)^n = (1 + \mathfrak{p}_k^s)^n \subseteq (k^\times)^n.$$

Exercise 4

Let K/k be finite abelian of exponent p. By local class field theory, for $a \in k^\times$ we have $(a^p, K/k) = (a, K/k)^p = \mathrm{id} \in \mathrm{Gal}(K/k)$. Hence $(k^\times)^p \subseteq \ker((\_, K/k))$. Conversely, if $(k^\times)^p \subseteq \ker((\_, K/k))$, then since the Artin map is surjective, $\mathrm{Gal}(K/k)$ has exponent (dividing) p. Hence the finite abelian extensions of exponent p correspond exactly to the subgroups $(k^\times)^p \subseteq H \subsetneq k^\times$ (they are automatically open).

By the same calculation as in exercise 3, for $p \neq 2$ we have $k^\times/(k^\times)^p \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$. This has $\frac{p^2-1}{p-1} = p + 1$ subgroups of order p, and the trivial subgroup, so there are $p + 2$ finite abelian extensions of $\mathbb{Q}_p$ of exponent p. For $p = 2$, the same method yields $k^\times/(k^\times)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$, which has 7 subgroups of order 2, and $\frac{1}{3}\binom{7}{2} = 7$ subgroups of order 4, so there are $1 + 7 + 7 = 15$ finite abelian extensions of $\mathbb{Q}_2$ of exponent 2.

Exercise 1

Since $k^\times \subseteq J_k$ are Hausdorff topological groups, it suffices to show that $\{1\} \subseteq k^\times$ is open, i.e. that there exists an open neighbourhood $1 \in U \subseteq J_k$ with $U \cap k^\times = \{1\}$. Let S_∞ be the set of real places of k, and set $U_\varepsilon := \prod_{v \in S_\infty} B_\varepsilon(1) \times \prod_{v \notin S_\infty} \mathcal{U}_v$ for $\varepsilon > 0$. By definition of the product topology, U_ε is open. Now let $\alpha \in k^\times$. Then $\alpha \in U_\varepsilon$ implies $\alpha \in \mathcal{O}_{k_v}$ for the finite places, i.e. $\alpha \in \mathcal{O}_k$, and $|\alpha - 1|_v < \varepsilon$ for every infinite place $v \in S_\infty$. But $\mathcal{O}_k$ embeds as a lattice in $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, which is in particular discrete. So for ε small enough, no $\alpha \in \mathcal{O}_k$ satisfies these conditions.

Exercise 2

Let cl_k be generated by the classes of $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. Then set $S := S_\infty \cup \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$. Let $\alpha \in J_k$ and denote by $\varphi : J_k/k^\times\mathcal{U}_k \to \mathrm{cl}_k$ the isomorphism from exercise 3. Then by construction, write $\varphi(\alpha) = x \prod \mathfrak{p}_i^{e_i}$ for some $e_i \in \mathbb{Z}$ and $x \in K^\times$. Set $\beta = \alpha x^{-1}$, we have to show $\beta \in J_k^S$. So let $v \notin S$, then v is finite. We have $v(\alpha) = v(\varphi(\alpha)) = v(x)$ since $v \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$, but $\varphi(\alpha)$ and x only differ by factors of the $\mathfrak{p}_i$. Hence $v(\beta) = v(\alpha) - v(x) = 0$ and $\beta \in \mathcal{U}_v$, as required.

From $J_k = J_k^S k^\times$ it immediately follows that $C_k^S = J_k^S k^\times / k^\times = J_k/k^\times = C_k$.

Exercise 3

(a) Consider the content map $\overline{c} : J_k \xrightarrow{c} I_k \twoheadrightarrow \mathrm{cl}_k$. The kernel of $\overline{c}$ is $c^{-1}(k^\times) = k^\times \ker c = k^\times \mathcal{U}_k$.

(b) Consider the commutative diagram defining the Artin map on J_k (after theorem 1.45).Let $\mathfrak{m} = 1$ and $K = k(1)$. We know that ideal-theoretically, K corresponds to $P_k(\mathfrak{m})$. Also, the top arrow is induced by the content, so it is exactly the map in (a). Finally, since $\mathfrak{m} = 1$ the left vertical arrow is the identity. Under all these simplifications, the Artin map reduces to the composition

$$J_k \twoheadrightarrow J_k/k^\times\mathcal{U}_k \xrightarrow[\cong]{\overline{c}} \mathrm{cl}_k \xrightarrow[\cong]{(\_, K/k)} \mathrm{Gal}(K/k).$$

Since the last two arrows are isomorphisms, the kernel is just the kernel of the first one, which is obviously $k^\times \mathcal{U}_k$.

Exercise 4

(a) It's clear that $\mathfrak{p}$ ramifies in L/k if and only if $\sigma(\mathfrak{p})$ ramifies in $\widehat{\sigma}(L)/k$. So $\widehat{\sigma}(L)/k$ is unramified outside $\sigma(\mathfrak{m})$, and the Artin map is well-defined on $I_k(\sigma(\mathfrak{m}))$. Now let $\mathfrak{p} \in I_k(\mathfrak{m})$ be a prime ideal, and $\tau := (\mathfrak{p}, L/k) \in \mathrm{Gal}(L/k)$ its Frobenius. Then τ is uniquely determined by $\tau(\alpha) \equiv \alpha^f \bmod \mathfrak{p}$, where $f := \mathrm{N}(\mathfrak{p})$. Let $\alpha \in \widehat{\sigma}(L)$, then $\tau(\widehat{\sigma}^{-1}\alpha) \equiv (\widehat{\sigma}^{-1}\alpha)^f \equiv \widehat{\sigma}^{-1}(\alpha^f) \bmod \mathfrak{p}$, hence $\widehat{\sigma}\tau\widehat{\sigma}^{-1}(\alpha) \equiv \widehat{\sigma}\widehat{\sigma}^{-1}(\alpha^f) \equiv \alpha^f \bmod \sigma(\mathfrak{p})$, which precisely means $(\sigma(\mathfrak{p}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{p}, L/K)\widehat{\sigma}^{-1}$. Since the prime ideals generate the fractional ideals, this identity automatically holds for all fractional ideals.

In particular, using that L is defined mod $\mathfrak{m}$, i.e. $P_k(\mathfrak{m}) \subseteq \ker(\_, L/k)$, we see for $\sigma(\mathfrak{a}) \in \sigma(P_k(\mathfrak{m})) = P_k(\sigma(\mathfrak{m}))$ that $(\sigma(\mathfrak{a}), \widehat{\sigma}(L)/k) = \widehat{\sigma}(\mathfrak{a}, L/k)\widehat{\sigma}^{-1} = \widehat{\sigma}\widehat{\sigma}^{-1} = 1$, i.e. $P_k(\sigma(\mathfrak{m})) \subseteq \ker(\_, \widehat{\sigma}(L)/k)$, as desired.

(b) Since $\ker(\_, \widehat{\sigma}(k_H)/k) = \sigma(\ker(\_, k_H/k))$, we see that $\widehat{\sigma}(k_H)$ corresponds to $\sigma(H)$.

Now k_H/F is galois if and only if $\widehat{\sigma}(k_H) = k_H$ for all $\widehat{\sigma}$, if and only if $\sigma(H) = H$, for all $\sigma \in G$, if and only if $\sigma(H) \subseteq H$ for all $\sigma \in G$.

Exercise 1

Let $\mathfrak{m} := \mathfrak{q}$ be a prime of k. By exercise 1.28, $[k(\mathfrak{m}) : H] = \frac{\omega(1)}{\omega(\mathfrak{m})}\varphi_k(\mathfrak{m}) = \frac{\omega(1)}{\omega(\mathfrak{m})}(\mathrm{N}(\mathfrak{q})-1)$. If $\mathfrak{q}$ is split in $k/\mathbb{Q}$, then $\mathrm{N}(\mathfrak{q}) =: q$ is a prime number, and by class field theory there exists a field $H \subseteq K \subseteq k(\mathfrak{m})$ with the desired properties (the conductor must divide $\mathfrak{q}$ and cannot be trivial) if and only if $p \mid [k(\mathfrak{m}) : H]$. Hence it suffices to show that there are infinitely many primes $q \equiv 1 \bmod p$ that split in k.

If $p = 2$ this is clear, so now let p be odd and $k = \mathbb{Q}(\sqrt{d})$, $d < 0$. q splits in k if and only if $1 = \left(\frac{d}{q}\right)$. Write $d = p^n e$ with $p \nmid e$, then by quadratic reciprocity, q splits in k if and only if $\left(\frac{e}{q}\right) = (-1)^{n(p-1)(q-1)/4}$, which is a congruence condition $\bmod 4e$. By the Chinese Remainder Theorem and Dirichlet's Theorem on Arithmetic Progressions, there are infinitely many primes $q \equiv 1 \bmod p$ satisfying this condition, hence infinitely many field extensions as required.

Exercise 2

Clearly it suffices to prove (b). Recall that I_G is the free abelian group generated by $\sigma - 1$, $1 \neq \sigma \in G$. Hence set $\varphi : I_G \to G/G'$, $\sigma - 1 \mapsto \sigma$ and extend linearly. Now

$$\varphi((\sigma-1)(\tau-1)) = \varphi((\sigma\tau-1) - (\sigma-1) - (\tau-1)) = \sigma\tau\sigma^{-1}\tau^{-1}G' = 1,$$

so $I_G^2 \subseteq \ker\varphi$ and we get a well-defined morphismn $\overline{\varphi} : I_G/I_G^2 \to G/G'$.

Conversely, let $\psi : G \to I_G/I_G^2$, $\sigma \mapsto \sigma - 1 + I_G^2$. For $\sigma, \tau \in G$, we have

$$\psi(\sigma\tau) = \sigma\tau - 1 + I_G^2 = (\sigma-1) + (\tau-1) + (\sigma-1)(\tau-1) = \psi(\sigma) + \psi(\tau).$$

Thus ψ is a group homomorphism. Since I_G/I_G^2 is abelian, it factors through $\overline{\psi} : G/G' \to I_G/I_G^2$. Clearly $\overline{\varphi}$ and $\overline{\psi}$ are mututally inverse, hence $I_G/I_G^2 \cong G/G' = G^{\mathrm{ab}}$.

Exercise 3

Let $G = \{\pm 1\}$, $A = \mathbb{Z}/2\mathbb{Z}$ with the trivial G-action and $B = \mathbb{Z}/4\mathbb{Z}$ with the action $-1 \cdot b = -b = 3b$. Then $B^G = \{0, 2\} \cong \mathbb{Z}/2\mathbb{Z}$, and $A \otimes B^G \cong (Z/2\mathbb{Z}) \otimes (\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ (recall that, in general, multiplication induces an isomorphism $C_n \otimes C_m \cong C_{\gcd(n,m)}$) with the non-trivial element $1 \otimes 2$. Its image in $A \otimes B = (\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/4\mathbb{Z})$ is $1 \otimes 2 = 2(1 \otimes 1) = 2 \otimes 1 = 0$, so the canonical map $A^G \otimes B^G \to (A \otimes B)^G$ is the zero map. Since $1 \otimes -1 = 3(1 \otimes 1) = 1 \otimes 1$, we have $(A \otimes B)^G = A \otimes B \cong \mathbb{Z}/2\mathbb{Z}$, so the map is not surjective either, even if the action of G on A is trivial.

Exercise 4

(1)⇒(4): Let $0 \to A \to B \to X \to 0$ be exact, with X projective. Thus there exists a morphism making the following diagram commute:

$$\begin{tikzcd} & X \arrow[dl, dashed, "\exists s"'] \arrow[d, "\mathrm{id}"] & \\ B \arrow[r] & X \arrow[r] & 0 \end{tikzcd}$$

which is exactly the required splitting.

(4)⇒(3): The morphism $\pi : B := \bigoplus_{x \in X} \mathbb{Z}[G] \to X$, $x \mapsto x$ is clearly surjective. By assumption let $s : X \to A$ be a splitting of the short exact sequence $0 \to \ker\pi \to B \to X \to 0$, then define $\varphi : \ker\pi \oplus X \to B$, $(a, x) \mapsto a + s(x)$. Let $(a, x) \in \ker\varphi$, then $0 = \pi(\varphi(a, x)) = \pi(\sigma(x)) = x$, hence also $a = \varphi(a, 0) = 0$. Conversely, let $b \in B$. Then $\pi(b - \sigma(\pi(b))) = 0$, so $b - \sigma(\pi(b)) = a$ for some $a \in \ker \pi$. But that exactly means $b = \varphi(a, \sigma(\pi(b)))$, so φ is an isomorphism and X is direct summand of a free G-module.

(3)⇒(2): Let $X \oplus Y = Z$ with Z a free G-Module. The result follows immediately from $\mathrm{Hom}(Z, \_)$ being exact and $\mathrm{Hom}(Z, \_) = \mathrm{Hom}(X, \_) \times \mathrm{Hom}(Y, \_)$.

(2)⇒(1): Let $g : B \to C$ be surjective, and $h : X \to C$ any morphism. Then by assumption $\mathrm{Hom}(X, B) \to \mathrm{Hom}(X, C) \to 0$ is surjective, so there is a preimage $s : X \to B$ of h.